

Numerical Analysis of Double Pendulum

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1 Motivation:

Describe the problem you are trying to solve or the physical system you are investigating, including why it is interesting.

Oscillations are extremely widespread. The double pendulum system is a complex and fascinating problem in classical mechanics. This simple physical setting involves complicated coupled differential equations, providing rich material for the study of mechanics. This project aims to explore the double pendulum using Lagrangian mechanics and numerical methods, including identifying the normal modes and demonstrating some chaotic results by varying the initial conditions.

This project aims to investigate the double pendulum system through the following key steps:

- Derive the Lagrangian of the system to establish the equations of motion.
- Solve the equations of motion analytically (under small angle approximation, linear).
- Find the numerical solution using RK4, Euler's Method, compare with analytical solution.
- Identify the normal modes by analyzing the eigenvalues and eigenvectors of the linear system.
- Explore the Phase Space plots for analyzing the trajectories, fixed points, and chaos patterns.
- Create an animation of the double pendulum.

2 Theory:

Describe any mathematical and physical background for your project, including explanations for the equations you will be implementing. If you are comparing a numerical method to an analytic method, include the analytic expressions you'll be using for comparison here as well.

In order to solve the equations of motion and normal modes of this system in a more systematic way, linear algebra methods like characteristic polynomials, eigenvalues, eigenvectors are used here.

- Use of "Mass" M and "Spring constant" K matrices
 - "Mass" matrix is not exactly the mass but it describe the masses influencing the motion.
 - "Spring Constant" matrix is similar to the spring constant that describes the restoring forces in the system
- Normal Mode Analysis
 - The method focuses on finding the normal modes of the system where all the masses moves in the same frequencies by with different amplitudes.
- Eigenvalues problem
 - $(K - \omega^2 M) a = 0$ determines the normal mode condition, where a is the amplitude vector, and ω is the angular frequency.
- General solution
 - Once the eigenvalues and eigenvectors are determined, the motion of the entire system can be represented as a superposition of the two normal modes. The solution usually takes the form of a linear combination of eigenvectors, with each combination term multiplied by a trigonometric function of the corresponding frequency.

$$T = \frac{1}{2} (m_1 + m_2) L_1^2 \dot{\phi}_1^2 + m_2 L_1 L_2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_1 - \phi_2) + \frac{1}{2} m_2 L_2^2 \dot{\phi}_2^2$$

$$U(\phi_1, \phi_2) = (m_1 + m_2) g L_1 (1 - \cos \phi_1) + m_2 g L_2 (1 - \cos \phi_2)$$

by small angle approximation ($L\ddot{\phi} = -g \sin \phi = -g\phi$ $\cos \phi \approx 1 - \phi^2/2$), Taylor expansion first 2 terms, and small product, the solvable equation becomes:

$$T = \frac{1}{2} (m_1 + m_2) L_1^2 \dot{\phi}_1^2 + m_2 L_1 L_2 \dot{\phi}_1 \dot{\phi}_2 + \frac{1}{2} m_2 L_2^2 \dot{\phi}_2^2$$

$$U = \frac{1}{2} (m_1 + m_2) g L_1 \phi_1^2 + \frac{1}{2} m_2 g L_2 \phi_2^2$$

The **analytical solution motion of equations** of ϕ_1 and ϕ_2

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}_1} = \frac{\partial \mathcal{L}}{\partial \phi_1} \quad \text{or} \quad (m_1 + m_2) L_1^2 \ddot{\phi}_1 + m_2 L_1 L_2 \ddot{\phi}_2 = -(m_1 + m_2) g L_1 \phi_1$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}_2} = \frac{\partial \mathcal{L}}{\partial \phi_2} \quad \text{or} \quad m_2 L_1 L_2 \ddot{\phi}_1 + m_2 L_2^2 \ddot{\phi}_2 = -m_2 g L_2 \phi_2$$

These two equations for ϕ_1 and ϕ_2 can be rewritten as a single matrix equation

$$\mathbf{M} \ddot{\boldsymbol{\phi}} = -\mathbf{K} \boldsymbol{\phi}$$

Introduce the (2×1) column of coordinates

$$\boldsymbol{\phi} = \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix}$$

and the two (2×2) matrices

$$\mathbf{M} = \begin{bmatrix} (m_1 + m_2) L_1^2 & m_2 L_1 L_2 \\ m_2 L_1 L_2 & m_2 L_2^2 \end{bmatrix} \quad \text{and} \quad \mathbf{K} = \begin{bmatrix} (m_1 + m_2) g L_1 & 0 \\ 0 & m_2 g L_2 \end{bmatrix}$$

To simplify the calculation, assume the pendulum has equal masses $m_1 = m_2 = m$, and equal lengths, $L_1 = L_2 = L$

Recall that $\omega_0 = \sqrt{g/L}$, so the matrices are

$$\mathbf{M} = mL^2 \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{K} = mL^2 \begin{bmatrix} 2\omega_0^2 & 0 \\ 0 & \omega_0^2 \end{bmatrix}$$

And its eigenvalue equation is matrix $(\mathbf{K} - \omega^2 \mathbf{M})$. That is

$$(\mathbf{K} - \omega^2 \mathbf{M}) = mL^2 \begin{bmatrix} 2(\omega_0^2 - \omega^2) & -\omega^2 \\ -\omega^2 & (\omega_0^2 - \omega^2) \end{bmatrix}$$

The **normal frequencies are given by the condition** $\det(\mathbf{K} - \omega^2 \mathbf{M}) = 0$, which is

$$2(\omega_0^2 - \omega^2)^2 - \omega^4 = \omega^4 - 4\omega_0^2 \omega^2 + 2\omega_0^4 = 0$$

with the two solutions $\omega^2 = (2 \pm \sqrt{2})\omega_0^2$.

Therefore, the two normal frequencies are given by

$$\omega_1^2 = (2 - \sqrt{2})\omega_0^2 \quad \text{and} \quad \omega_2^2 = (2 + \sqrt{2})\omega_0^2$$

(or $\omega_1 \approx 0.77\omega_0$ and $\omega_2 \approx 1.85\omega_0$) where $\omega_0 = \sqrt{g/L}$ is the frequency of a single pendulum of length L .

By knowing the two normal frequencies, the motion of the double pendulum can be solved by $(\mathbf{K} - \omega^2 \mathbf{M}) \mathbf{a} = 0$

let $\omega = \omega_1$ and ω_2 in turn:

$$(\mathbf{K} - \omega_1^2 \mathbf{M}) = mL^2 \omega_0^2 (\sqrt{2} - 1) \begin{bmatrix} 2 & -\sqrt{2} \\ -\sqrt{2} & 1 \end{bmatrix}$$

$$(\mathbf{K} - \omega_2^2 \mathbf{M}) = -mL^2 \omega_0^2 (\sqrt{2} + 1) \begin{bmatrix} 2 & \sqrt{2} \\ \sqrt{2} & 1 \end{bmatrix}$$

As a result, the equation $(\mathbf{K} - \omega_1^2 \mathbf{M}) \mathbf{a} = 0$ implies that $a_2 = \sqrt{2}a_1$, write in the form of $a_1 = A_1 e^{-i\delta_1}$, the two coordinates are

$$\boldsymbol{\phi}(t) = \begin{bmatrix} \phi_1(t) \\ \phi_2(t) \end{bmatrix} = \text{Re} \mathbf{a} e^{i\omega_1 t} = A_1 \begin{bmatrix} 1 \\ \sqrt{2} \end{bmatrix} \cos(\omega_1 t - \delta_1) \quad \textbf{first mode}$$

The equation $(\mathbf{K} - \omega_2^2 \mathbf{M}) \mathbf{a} = 0$ implies that $a_2 = -\sqrt{2}a_1$, similarly, the two coordinates are

$$\boldsymbol{\phi}(t) = \begin{bmatrix} \phi_1(t) \\ \phi_2(t) \end{bmatrix} = \text{Re} \mathbf{a} e^{i\omega_2 t} = A_2 \begin{bmatrix} 1 \\ -\sqrt{2} \end{bmatrix} \cos(\omega_2 t - \delta_2) \quad \textbf{second mode}$$

Finally, the **general solution** of the system is the linear combination of these two normal modes solutions:

$$\begin{aligned} \boldsymbol{\phi}(t) &= \begin{bmatrix} \phi_1(t) \\ \phi_2(t) \end{bmatrix} = A_1 \begin{bmatrix} 1 \\ \sqrt{2} \end{bmatrix} \cos(\omega_1 t - \delta_1) + A_2 \begin{bmatrix} 1 \\ -\sqrt{2} \end{bmatrix} \cos(\omega_2 t - \delta_2) \\ \phi_1(t) &= A_1 \cos(\omega_1 t - \delta_1) + A_2 \cos(\omega_2 t - \delta_2) \\ \phi_2(t) &= A_1 \sqrt{2} \cos(\omega_1 t - \delta_1) - A_2 \sqrt{2} \cos(\omega_2 t - \delta_2) \end{aligned}$$

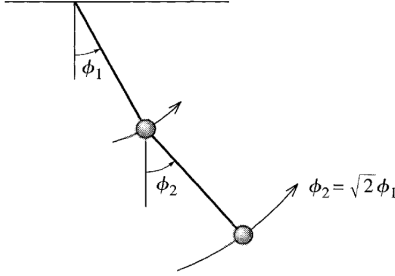


Figure 1: first normal mode, in phase. Taylor

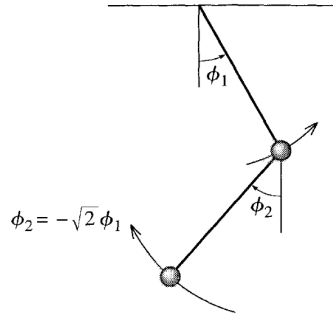


Figure 2: second normal mode, out of phase. Taylor

Notice that the:

$$\frac{\pi}{12} \text{rad is } 15^\circ$$

3 Methods:

The numerical methods utilized in this project are RK4 and Euler's methods.

Model Fitting: The project includes a comparison of the analytical model with the numerical methods, along with a comparison of the errors.

Data Visualization: Graphical representations of both the analytical and numerical results are provided. This includes phase space plots and animations. By examining the relationships between the angles and angular velocities, fascinating results are uncovered in the phase space, showing the system is sensitive.

The general solution for the small angle approximation of the double pendulum is determined.

The packages employed include `numpy`, `matplotlib.pyplot`, `matplotlib.animation`, and `IPython.display`, along with other subpackages of `matplotlib` that are detailed within the script.

This project is inspired by Chapter 11 of the Classical Mechanics by Taylor. The code and analysis are based on the lab textbook Computational Physics with Python

4 Results:

4.1 Trajectories

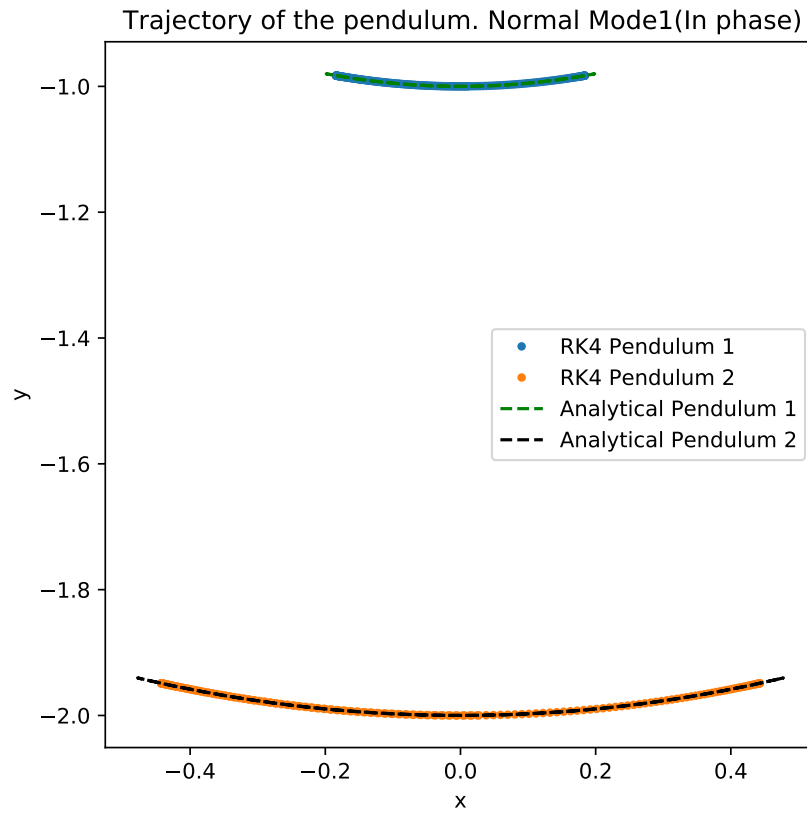


Figure 3: Mode 1 Trajectory. $t=2$ $N=100$

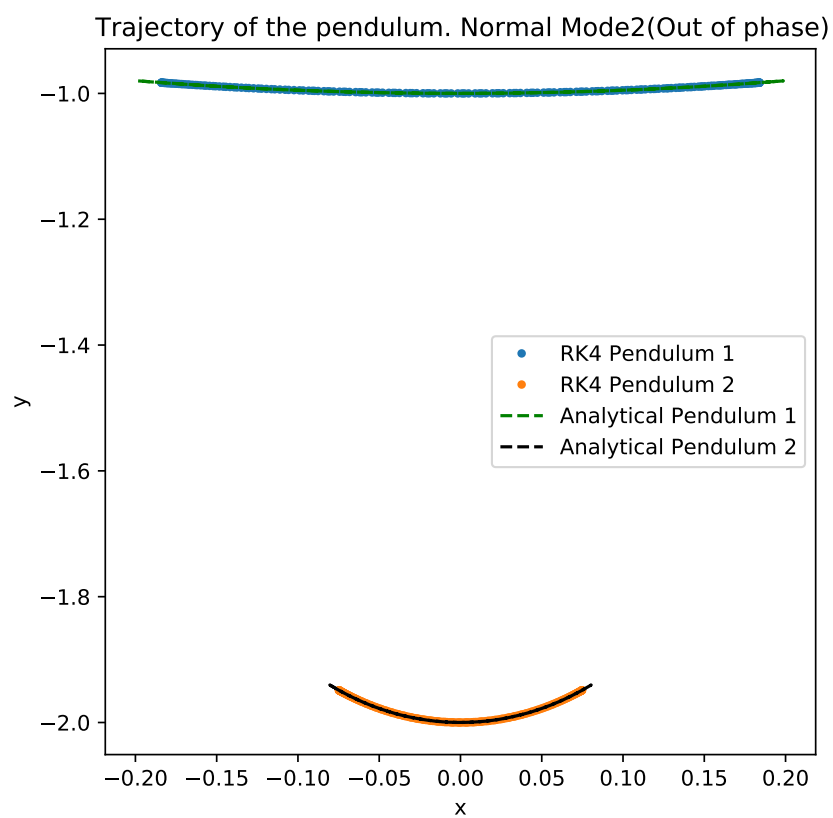


Figure 4: Mode 2 Trajectory. $t=2$ $N=100$

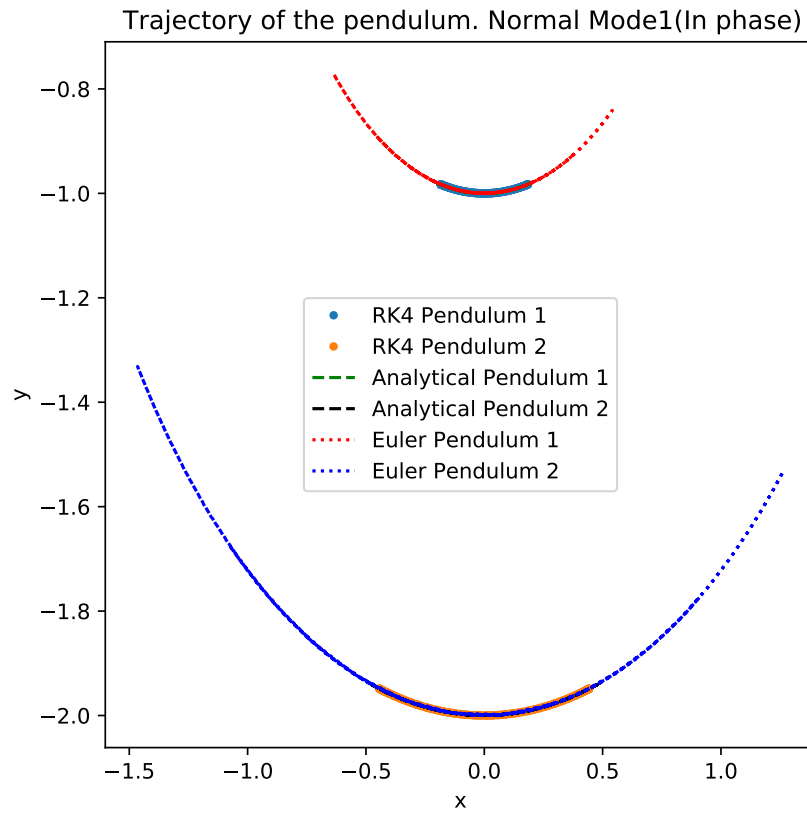


Figure 5: With Euler method in the comparison

As the plot shown, the Euler method does not fit well with the analytical solution.

4.2 Error Comparison

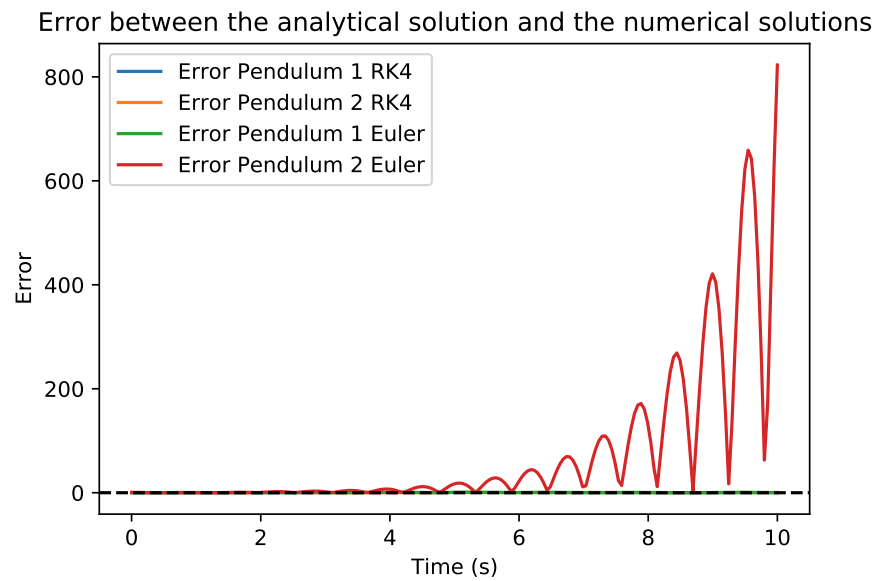


Figure 6: Comparison between numerical methods

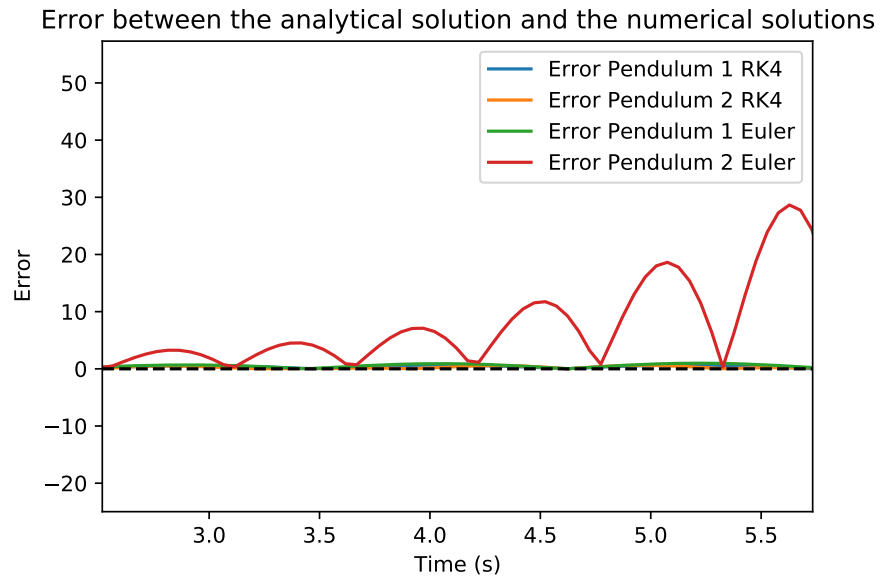


Figure 7: Zoom in

The plots show that Euler method has bigger errors compared with the RK4 method. The green and red lines are significantly higher than the zero, compared to the blue and orange line.

In phase out of phase April 22 explanation

4.3 Angle and Angular velocity with respect to Time

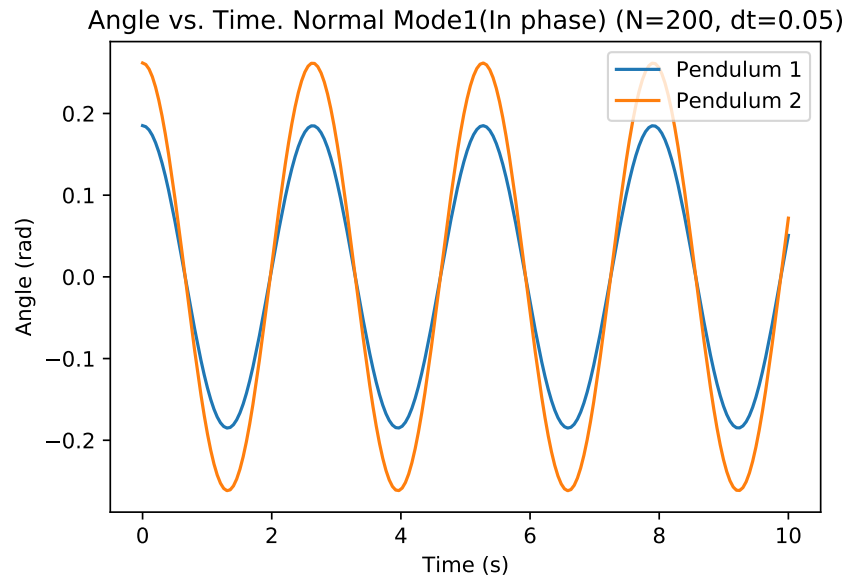


Figure 8: Angle vs. Time. Normal Mode 1

The RK4 numerical result matches the expectation that two pendulums are in phase and with the same frequency

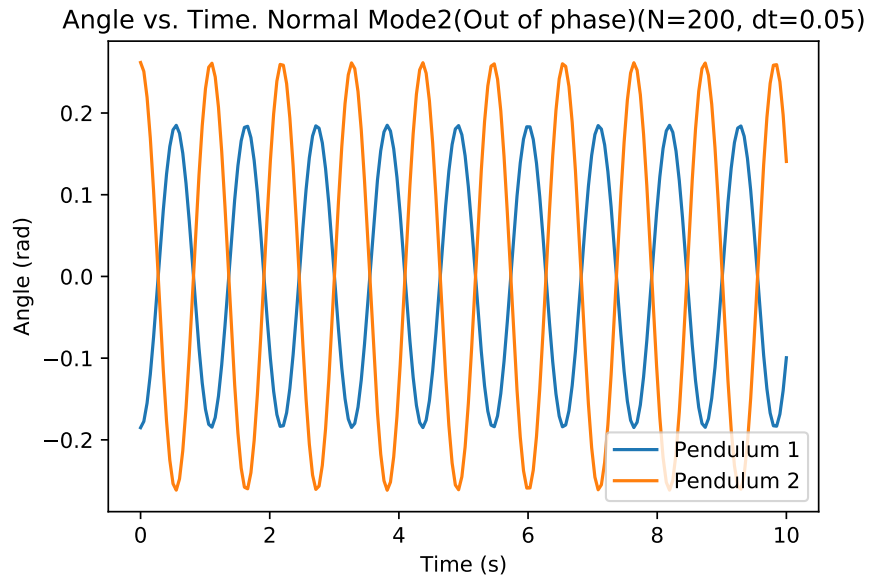


Figure 9: Angle vs. Time. Normal Mode 2

The RK4 numerical result matches the expectation that two pendulums are out of phase but with the same frequency.

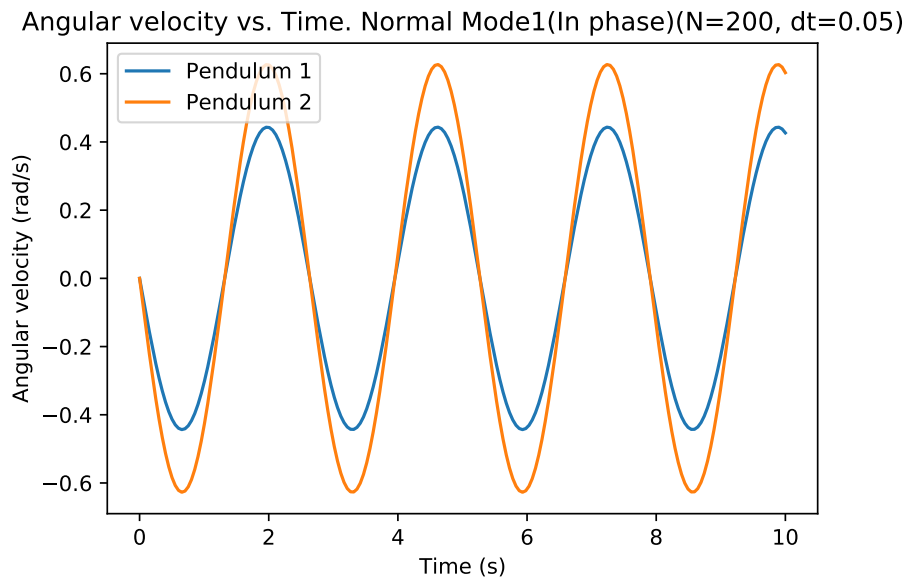


Figure 10: Angular velocity vs. time

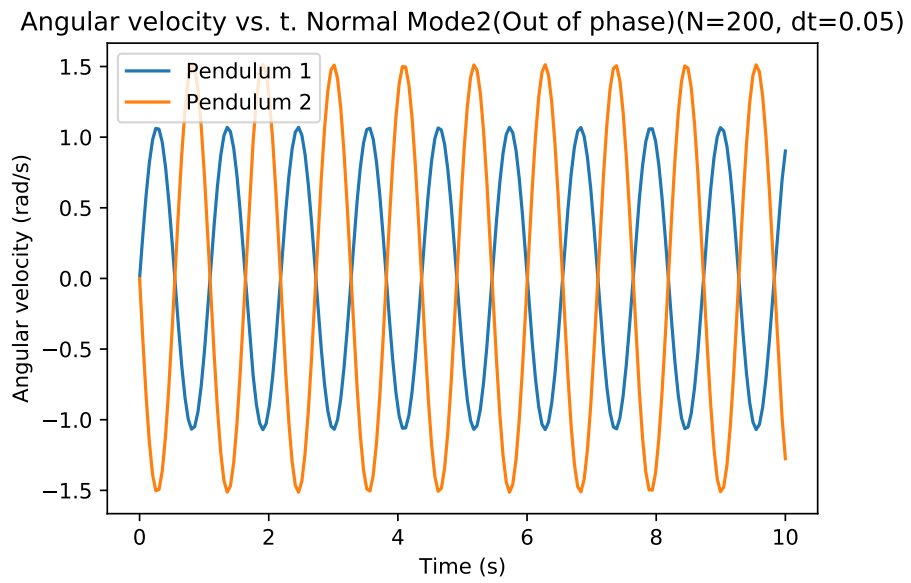


Figure 11: Angular velocity vs. time

4.4 Phase-Space Plot

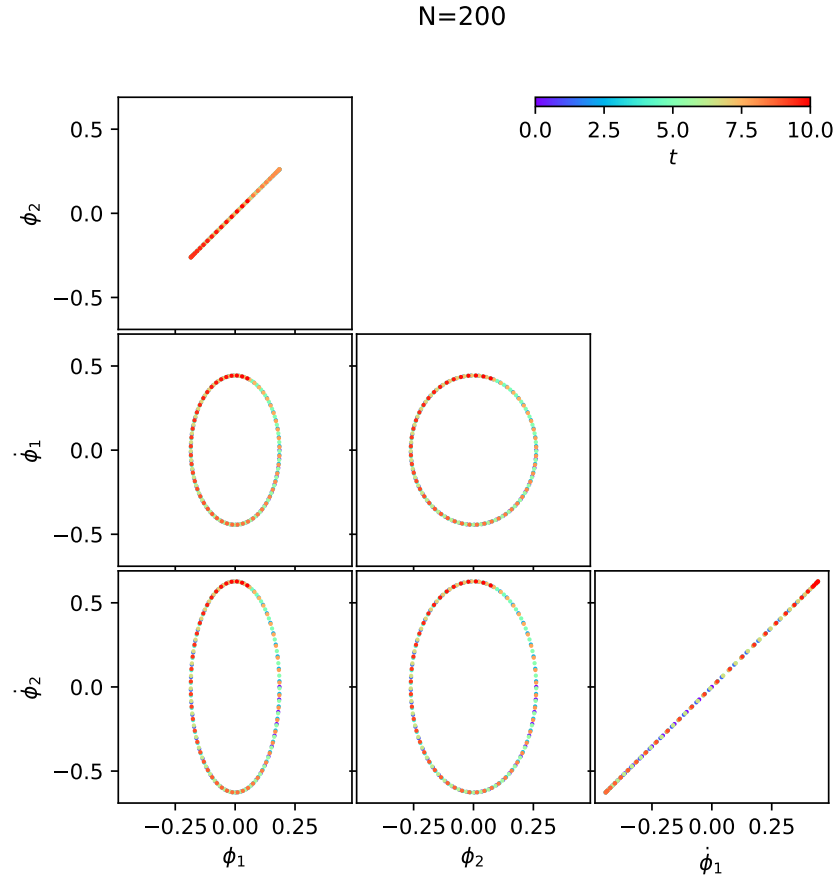


Figure 12: Normal Mode 1, in phase

N=200

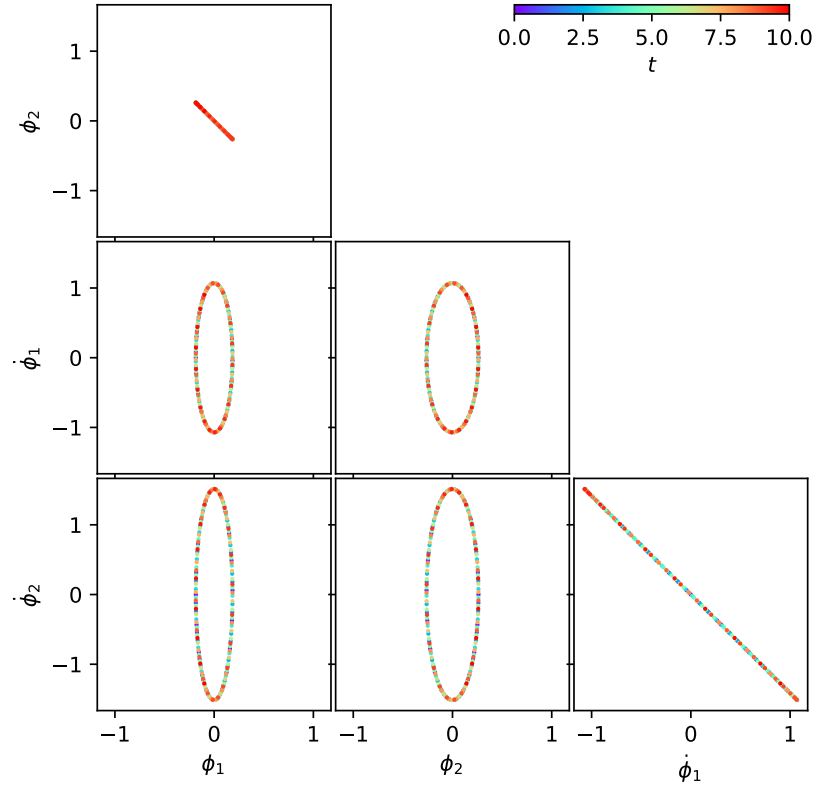


Figure 13: Normal Mode 2, out of phase

N=100

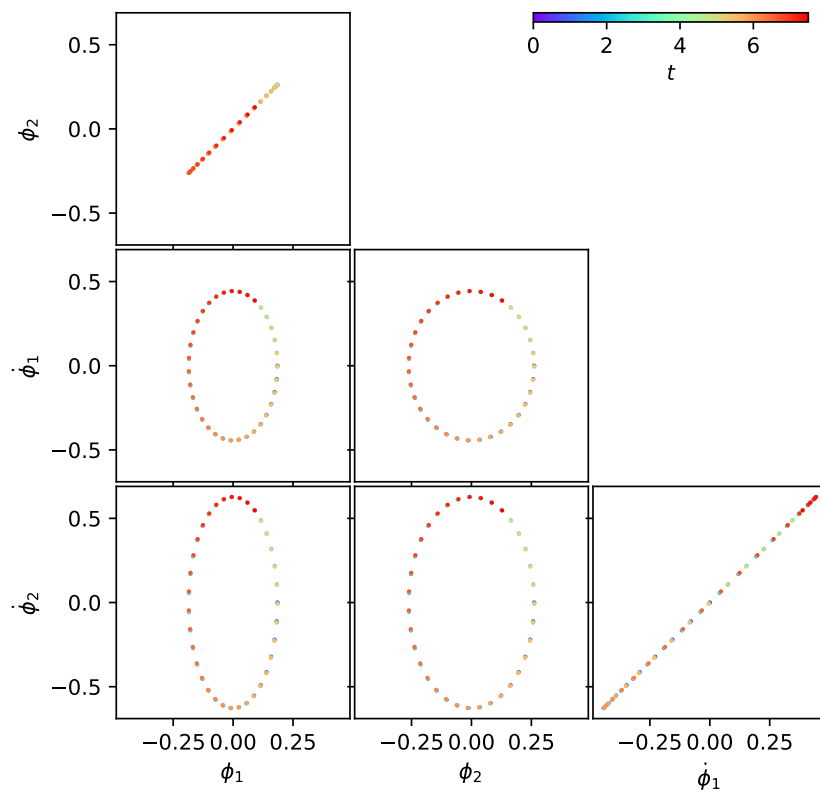


Figure 14: Normal Mode 1 $t=7.5s$

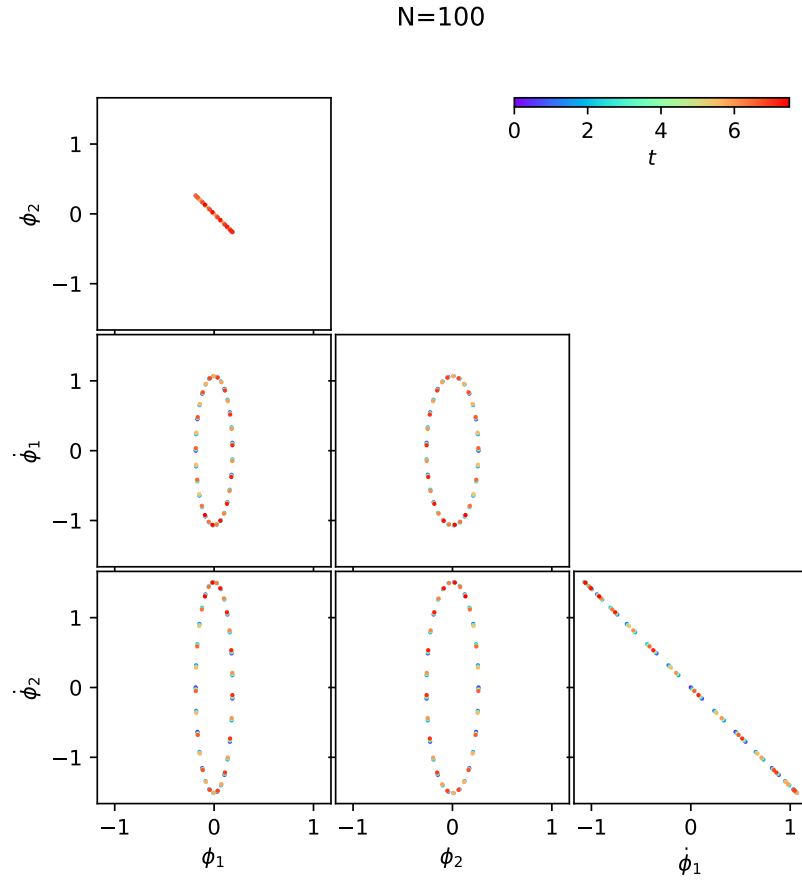


Figure 15: Normal Mode 2 $t=7.5$ s

Attractor is a special set of states to which a dynamical system tends to evolve over a long period of time. It reflects the asymptotic behavior of the system.

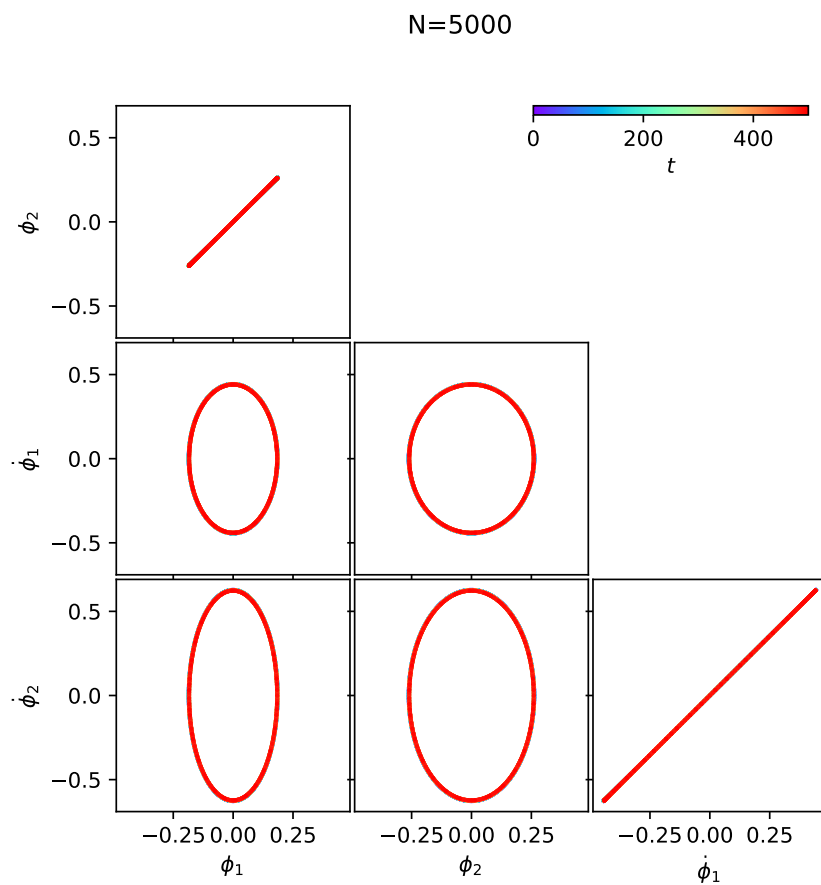


Figure 16: Attractor of mode 1

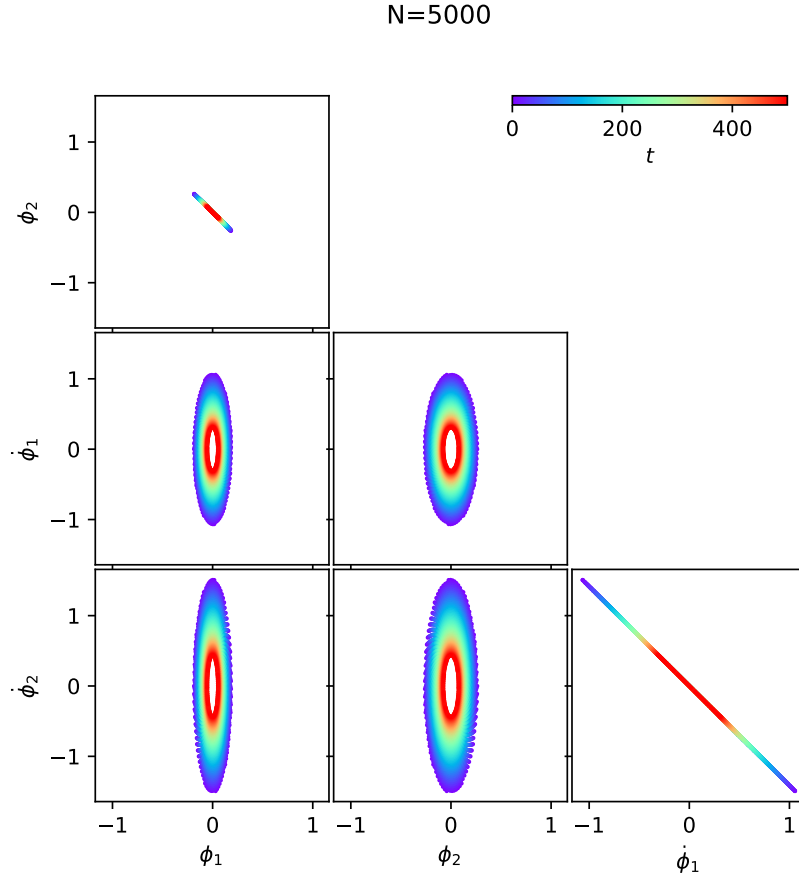


Figure 17: Attractor of mode 2

4.5 Animation

4.6 Various initial conditions

The system is very sensitive since little change of the initial conditions can lead to very different behaviors. This section provides some result by changing some initial condition.

5 Conclusions:

Clearly describe what your results demonstrate: did your numerical result agree with the analytic prediction (if relevant)? What interesting behaviors did you

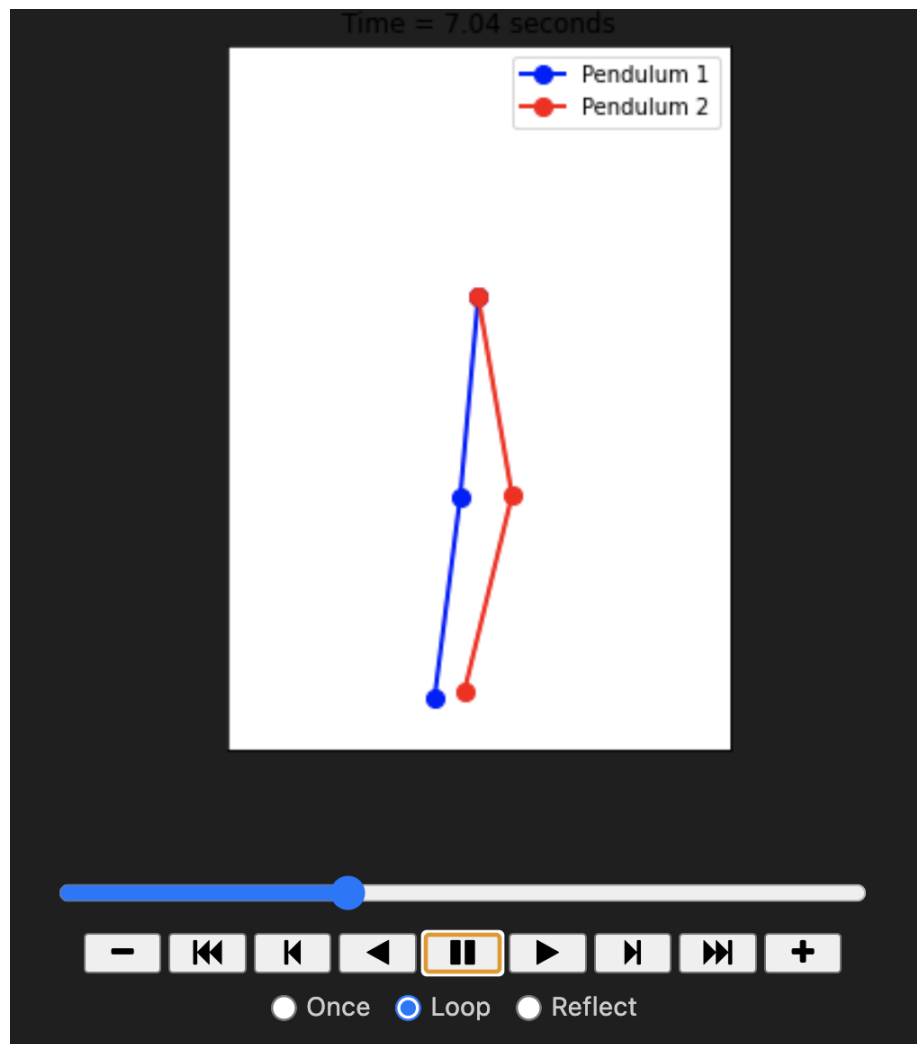


Figure 18: The animation UI is in the ipynb file

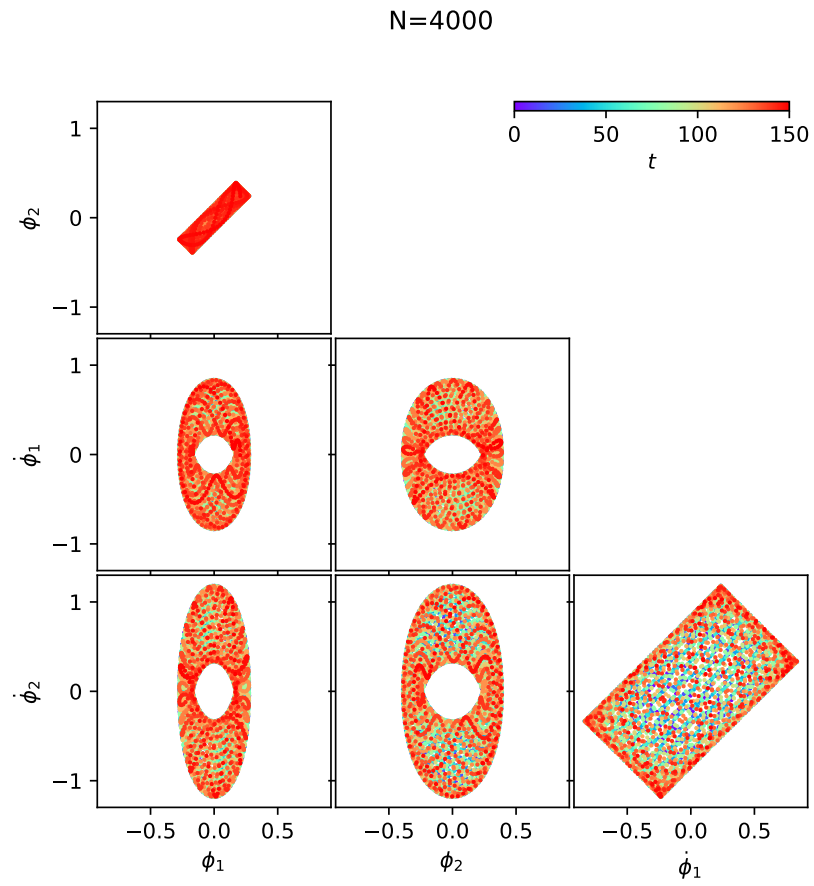


Figure 19: Give pendulum 1 a initial angular velocity 0.6 rad/s Normal Mode 1

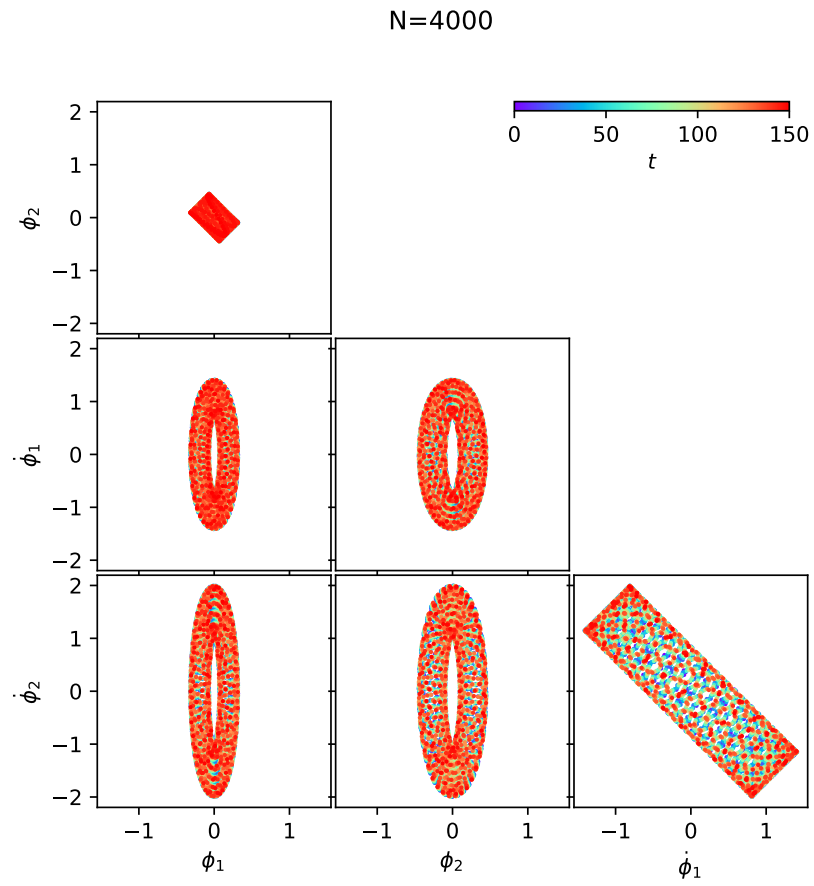


Figure 20: Give pendulum 1 a initial angular velocity 0.6 rad/s Normal Mode 2

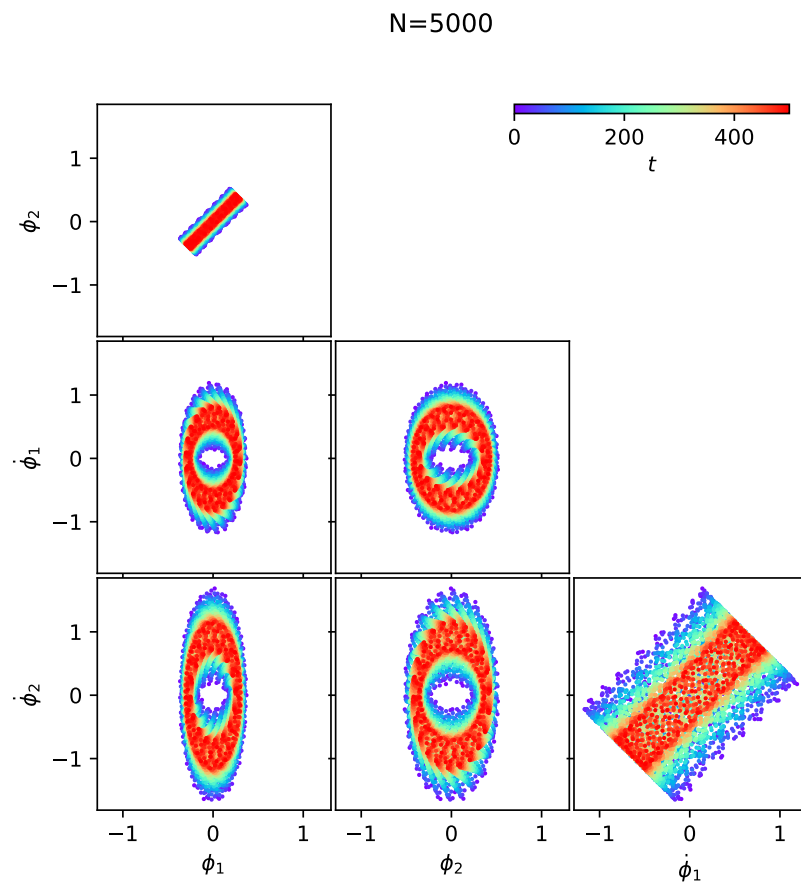


Figure 21: Change ϕ_1 to be $2\phi_1$ Normal Mode 1

observe? How does your project help you understand the behavior of the system or the solution to the problem you are investigating?

Trajectory Plots (Figures 4 and 5): These plots show the paths traced by the two pendulums over time. In Figure 4, which is the in-phase normal mode, the pendulums oscillate in unison. Figure 5 depicts the out-of-phase normal mode, where the pendulums swing in the opposite directions. The close agreement between the RK4 numerical solutions and the analytical solutions (dashed lines) indicate the accuracy of the numerical method.

Error Comparison (Figures 7 and 8): These plots quantify and visualize the errors between the analytical solutions and the numerical solutions obtained using RK4 and Euler methods. Figure 7 shows that the Euler method (green and red lines) accumulates significant errors over time compared to the RK4 method (blue and orange lines). Figure 8 zooms in to highlight the difference more clearly. This demonstrates the superiority of the RK4 method in terms of numerical accuracy and stability.

Angle and Angular Velocity vs. Time (Figures 9-12): These plots show how the angles and angular velocities of the pendulums evolve over time for the two normal modes. Figures 9 and 11 correspond to the in-phase mode, where the pendulums oscillate with the same frequency. Figures 10 and 12 illustrate the out-of-phase mode, with the pendulums oscillating at the same frequency but with a phase difference. The smooth sinusoidal curves indicate the periodic nature of the motion in the normal modes.

Phase Space Plots (Figures 13-18): The phase space plots provide a overview of the system's dynamics by plotting the angles and angular velocities with each other. Each plot consists of multiple subplots: the time evolution of the angles and angular velocities, and the phase portraits for each pendulum. Figures 13 and 14 show the phase space plots for the in-phase and out-of-phase normal modes over a longer time period. The closed orbits in the phase indicate the periodic nature of the motion. Figures 15 and 16 focus on a shorter time interval, showing the detailed structure of the orbits. Figures 17 and 18 show the longer time scale behavior, where the trajectories fill out regions of the phase space, and the presence of attractors.

Sensitivity to Initial Conditions (Figures 19-21): These plots show the sensitivity of the double pendulum system to small changes in the initial conditions. Figure 19 shows how giving pendulum 1 an initial angular velocity of 0.6 rad/s significantly changes the phase space portrait compared to the in-phase normal mode. Figure 20 demonstrates a similar effect for the out-of-phase normal mode. Figure 21 illustrates the impact of doubling the initial angle of pendulum 1. These results show the chaotic nature of the double pendulum, where slight modification can lead to dramatically different results over time.